

Q1.

A particle moves on a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

At time t , the particle's position vector, $\mathbf{r}$, is given by

$$\mathbf{r} = 4 \cos 3t \mathbf{i} - 4 \sin 3t \mathbf{j}$$

- (a) Prove that the particle is moving on a circle, which has its centre at the origin. (2)
- (b) Find an expression for the velocity of the particle at time t . (2)
- (c) Find an expression for the acceleration of the particle at time t . (2)
- (d) The acceleration of the particle can be written as

$$\mathbf{a} = k\mathbf{r}$$
 where k is a constant.
 Find the value of k . (2)
- (e) State the direction of the acceleration of the particle. (1)

(Total 9 marks)

Q2.

A particle moves on a horizontal plane in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

At time t seconds, the particle's position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 8 \begin{pmatrix} \cos \frac{1}{4}t \\ \sin \frac{1}{4}t \end{pmatrix} \mathbf{i} - 8 \begin{pmatrix} \sin \frac{1}{4}t \\ \cos \frac{1}{4}t \end{pmatrix} \mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t . (2)
- (b) Show that the speed of the particle is a constant. (3)
- (c) Prove that the particle is moving in a circle. (2)
- (d) Find the angular speed of the particle. (2)

- (e) Find an expression for the acceleration of the particle at time t .

(2)

- (f) State the magnitude of the acceleration of the particle.

(1)

(Total 12 marks)



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